

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		Kills bacteria	1			1		
		(ii)		(Gram positive) {have (thick) peptidoglycan in cell wall / no additional lipopolysaccharides} (so more susceptible to antibiotic) (1) - Inhibits formation of cross linkages between {peptidoglycans/ amino acids} (1) (peptidoglycan) cell wall is {broken down/ digested} /Cell wall is weakened (1) {Osmosis/ water moves in} causing cell to {lyse/ burst} (1)	1	2	1	4		1
		(iii)		Any three (×1) from: - Reference to mutation (1) (means some bacteria) become <u>resistant</u> (1) (The resistant bacteria) have a selective advantage /are able to survive antibiotic (1) - Resistance passed on (when bacteria divide/ reproduce/to next generation) (1)		3		3		
		(iv)		Don't scratch bites/ wash area/ use of anti-bacterial cream/ AVP Reject antibiotics/ insect repellents			1	1		
	(b)	(i)		Endemic – (always) present at low levels, epidemic – significant increase (in usual number of cases) (1) Pandemic – affects a large number of people/occurs {over a wide area/ many countries/ global} (1)	1		1	2		